

The combinatorial invariance conjecture

Grant Barkley
University of Michigan

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- 1 Kazhdan–Lusztig polynomials
- 2 The Bruhat graph
- 3 The Combinatorial Invariance Conjecture
- 4 Hypercube decompositions
- 5 Coefficient of q

Permutations and polynomials

A **permutation** is an ordering of the numbers $\{1, \dots, n\}$. The set of permutations is the **symmetric group** S_n .

Associated to any two permutations u and v is a polynomial $P_{u,v}(q)$

$$P_{123,321} = 1$$

$$P_{1324,3412} = 1 + q$$

$$P_{21,12} = 0$$

This is the **Kazhdan–Lusztig polynomial** of u, v

$P_{u,v} \neq 0$ if and only if $u \leq v$ in **Bruhat order**

Kazhdan–Lusztig polynomials

The Kazhdan–Lusztig polynomials $P_{u,v}(q)$ for $u, v \in S_n$ are important numerical invariants in geometry, algebra, and representation theory.

Singularities of Schubert varieties

Representations of $\mathfrak{gl}_n$

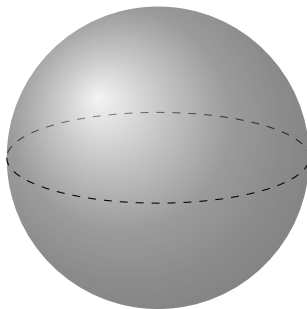
$$P_{u,v}(q)$$

Hecke algebra of S_n

Motivation 1: Singularities of Schubert varieties

For each $v \in S_n$, there is a **Schubert variety** $\overline{X}_v$

$$\overline{X}_{2134} = \mathbb{P}^1$$

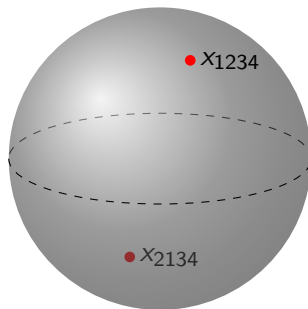


Motivation 1: Singularities of Schubert varieties

For each $v \in S_n$, there is a **Schubert variety** $\overline{X}_v$

For certain $u \in S_n$, there is an associated point $x_u \in \overline{X}_v$

$$\overline{X}_{2134} = \mathbb{P}^1$$



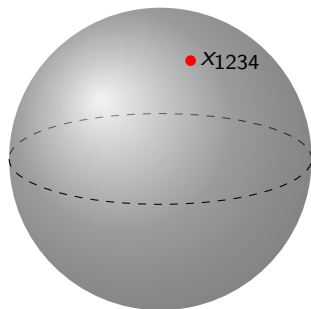
The point $x_u \in \overline{X}_v$ if and only if $u \leq v$ in **Bruhat order**

Motivation 1: Singularities of Schubert varieties

$P_{u,v}(q)$ describes the singularities of $\overline{X}_v$ near the point x_u

Example: $P_{u,v}(q) = 1$ if and only if $\overline{X}_v$ is smooth at x_u

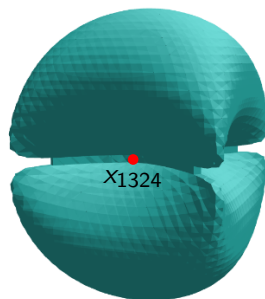
$\overline{X}_{2134}$



x_{1234}

$$P_{1234,2134} = 1$$

$\overline{X}_{3412}$



x_{1324}

$$P_{1324,3412} = 1 + q$$

Motivation 2: Representations of $\mathfrak{gl}_n$

Let $\lambda = (\lambda_1, \dots, \lambda_n)$ and $\mu = (\mu_1, \dots, \mu_n)$ be sequences of n integers

Associated to λ and μ is a vector space V_λ^μ from representation theory:

“The μ -weight space of the irreducible $\mathfrak{gl}_n$ -representation of highest weight λ ”

The **character** of V_λ is $\text{ch}(V_\lambda) = \sum_\mu \dim(V_\lambda^\mu) \mathbf{x}^\mu$

If λ is a partition (decreasing sequence of non-negative integers), then $\text{ch}(V_\lambda)$ is a polynomial (a **Schur polynomial**)

Motivation 2: Representations of $\mathfrak{gl}_n$

$$\rho = (n, n-1, \dots, 2, 1)$$

Weyl character formula: if λ is a partition then

$$\text{ch}(V_\lambda) = \frac{\sum_{w \in S_n} (-1)^w \mathbf{x}^{w(\lambda+\rho)-\rho}}{\prod_{\alpha \in \Delta^+} (1 - \mathbf{x}^{-\alpha})}$$

The **Kazhdan–Lusztig conjecture**: if $v(\lambda + \rho)$ is a partition* then

$$\text{ch}(V_\lambda) = \frac{\sum_{w \in S_n} (-1)^w P_{ww_0, v w_0}(1) \mathbf{x}^{w(\lambda+\rho)-\rho}}{\prod_{\alpha \in \Delta^+} (1 - \mathbf{x}^{-\alpha})}$$

*: with all entries distinct

What is a KL polynomial?

What is a KL polynomial?

Motivation 1: the Poincaré polynomial of the stalk of an **intersection cohomology** complex on $\overline{X}_v$

$$P_{u,v}(q) = \sum_i q^i \dim H^{-\ell(u,v)+2i} \mathbf{IC}(\overline{X}_v)_{x_u}$$

Motivation 2: the Poincaré polynomial of the $R\mathrm{Hom}$ complex between Verma and simple $\mathfrak{gl}_n$ -modules

$$P_{u,v}(q) = \sum_i q^i \dim \mathrm{Ext}^{\ell(u,v)-2i}(M_u, L_v)$$

Motivation 3: transition matrix for canonical basis of Hecke algebra

$$C_v = q^{-\frac{\ell(v)}{2}} \sum_{u \in S_n} P_{u,v}(q) T_u$$

What is a KL polynomial?

The **Kazhdan–Lusztig polynomials** are the unique family of polynomials $P_{u,v}(q) \in \mathbb{Z}[q]$ satisfying:

- (1) $P_{u,v}(q) = 0$ unless $u \leq v$.
- (2) $P_{u,u}(q) = 1$ for all u .
- (3) $\deg P_{u,v}(q) < \frac{1}{2}(\ell(v) - \ell(u))$ if $u < v$.
- (4) And, if $u < v$ then

$$q^{\ell(u,v)} P_{u,v} \left(\frac{1}{q} \right) - P_{u,v}(q) = \sum_{x \in (u,v]} R_{u,x}(q) P_{x,v}(q).$$

What is an R -polynomial?

One answer:

Associated to u and v is a **Richardson variety** $\mathcal{R}_{u,v}$. Then

$$R_{u,v}(q) = |\mathcal{R}_{u,v}(\mathbb{F}_q)|.$$

What is an R -polynomial?

The **R -polynomials** are the unique family of polynomials $R_{u,v}(q) \in \mathbb{Z}[q]$ satisfying:

- (1) $R_{u,v}(q) = 0$ unless $u \leq v$.
- (2) $R_{u,u}(q) = 1$ for all u .
- (3) If $us < u$ and $vs < v$:

$$R_{u,v}(q) = R_{us,vs}(q).$$

- (4) If $us > u$ and $vs < v$:

$$R_{u,v}(q) = qR_{us,vs}(q) + (q - 1)R_{u,vs}(q).$$

In summary...

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- You can compute $P_{u,v}$
- But, it's complicated
- First you need to compute $R_{u,v}$
- And that's also complicated (big recursion)
- But the end result encodes important numerical invariants from different fields

Mysteries about Kazhdan–Lusztig polynomials

Kazhdan–Lusztig conjecture

Why is $P_{u,v}$ in representation theory?

Non-negativity conjecture

Why are the coefficients of $P_{u,v}$ always positive?

Combinatorial invariance conjecture

Why is $P_{u,v}$ determined by the Bruhat graph?

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Kazhdan–Lusztig conjecture

Solved in 1981

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Unsolved!

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Bruhat order

We say $u \leq v$ in **Bruhat order** if and only if we can transform u into v by swapping pairs of entries that are in increasing order

$$123 \leq 312$$

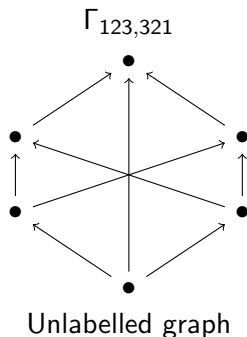
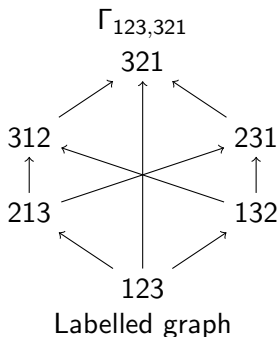
$$123 \rightarrow 213 \quad 213 \rightarrow 312$$

$$132 \rightarrow 312 \quad 312 \rightarrow 321 \quad 132 \not\leq 213 \quad \text{or} \quad 132 \rightarrow 231 \quad 231 \rightarrow 321$$

Bruhat graphs

The **Bruhat graph** $\Gamma_{u,v}$ is a directed graph

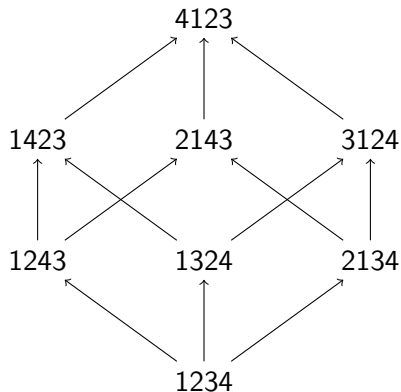
- Vertex set: Vertices are permutations x so that $u \leq x$ and $x \leq v$
- Edges: There is an edge $x \rightarrow y$ if we can transform x into y by swapping two increasing entries



We write $\ell(u, v)$ for the maximal length of a path from u to v

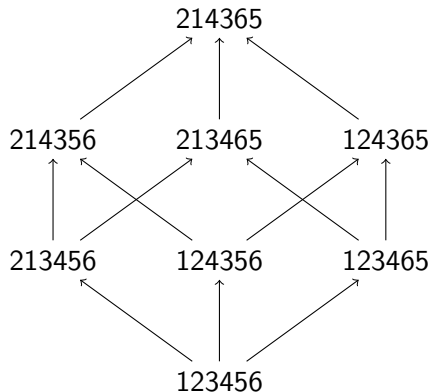
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Bruhat graphs and polynomials



$$R_{1234,4123} = (q - 1)^3$$

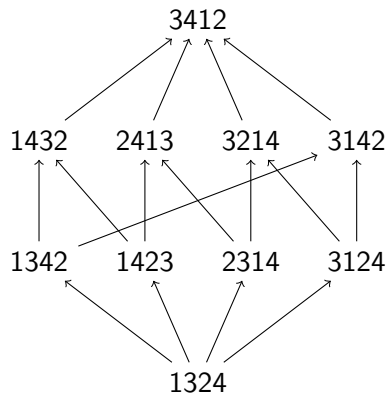
$$P_{1234,4123} = 1$$



$$R_{123456,214365} = (q - 1)^3$$

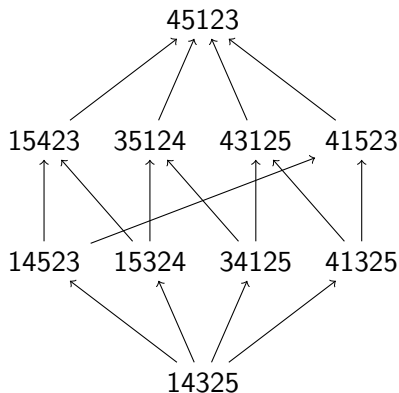
$$P_{123456,214365} = 1$$

Bruhat graphs and polynomials



$$R_{1324,3412} = (q - 1)^3$$

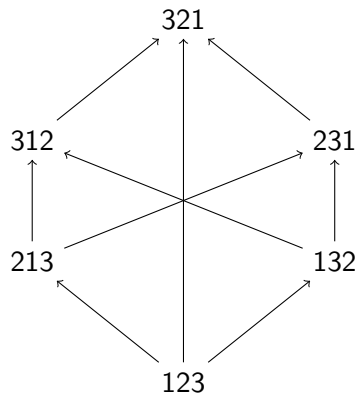
$$P_{1324,3412} = 1 + q$$



$$R_{14325,45123} = (q - 1)^3$$

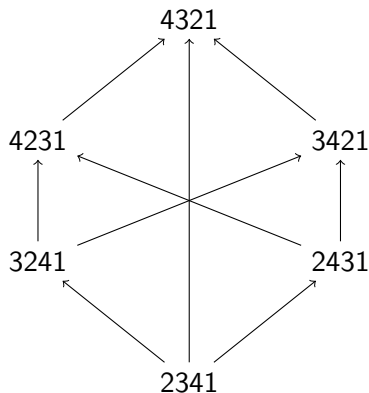
$$P_{14325,45123} = 1 + q$$

Bruhat graphs and polynomials



$$R_{123,321} = (q-1)^3 + q(q-1)$$

$$P_{123,321} = 1$$



$$R_{2341,4321} = (q-1)^3 + q(q-1)$$

$$P_{2341,4321} = 1$$

Combinatorial Invariance Conjecture

Lusztig and Dyer made the following surprising conjecture:

Conjecture (Lusztig 1980s; Dyer 1987)

If $\Gamma_{u,v}$ and $\Gamma_{u',v'}$ are isomorphic unlabelled directed graphs, then

$$P_{u,v}(q) = P_{u',v'}(q) \quad \text{and} \quad R_{u,v}(q) = R_{u',v'}(q).$$

- Jantzen 1979, Dyer 1987: true when $P_{u,v}(q) = 1$.
- Brenti 1994: true when all edges of $\Gamma_{u,v}$ have length 1.
- Dyer 1987, Incitti 2006 & 2007: true when $\ell(u, v) \leq 8$.
- Brenti, du Cloux, Delanoy, Brenti–Caselli–Marietti 2003–2006: true when $u = u' = e = 123 \cdots n$.
- Patimo 2021: coefficients of q are equal.

Combinatorial Invariance Conjecture

Most of these results in fact prove:

Strong Combinatorial Invariance Conjecture

Given $\Gamma_{u,v}$ as an unlabelled directed graph, there is an algorithm for computing $P_{u,v}(q)$ from $\Gamma_{u,v}$.

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- Patimo 2021: coefficients of q are equal.

Examples of combinatorial invariance

Theorem (Brenti 1994)

The polynomial $R_{u,v}(q)$ factors as $(q-1)^{\ell(u,v)}$ if and only if all edges of $\Gamma_{u,v}$ have length 1.

General fact:

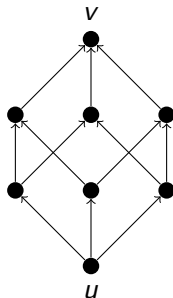
If we know $R_{x,y}$ for every $x, y \in \Gamma_{u,v}$, then we can also compute $P_{u,v}!$

Examples of combinatorial invariance



$$R_{u,v}(q) = q - 1$$
$$P_{u,v}(q) = 1$$

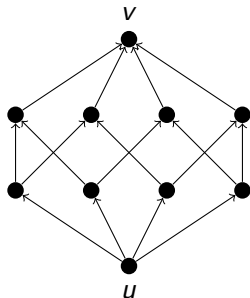
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Examples of combinatorial invariance



$$R_{u,v}(q) = (q - 1)^3$$

$$P_{u,v}(q) = 1 + q$$

A new combinatorial invariance theorem

Recall $e = 123 \cdots n$

Theorem (Brenti, du Cloux 2003)

Suppose $\Gamma_{e,v}$ and $\Gamma_{e,v'}$ are isomorphic. Then $R_{e,v}(q) = R_{e,v'}(q)$.

Theorem (B.–Gaetz 2023)

Suppose $\Gamma_{e,v}$ and $\Gamma_{u',v'}$ are isomorphic. Then $R_{e,v}(q) = R_{u',v'}(q)$.

A more general combinatorial invariance theorem

Let $[q]P_{u,v}$ be the coefficient of q in $P_{u,v}$.

Theorem (Patimo 2021)

If $\Gamma_{u,v}$ and $\Gamma_{u',v'}$ are isomorphic, then $[q]P_{u,v} = [q]P_{u',v'}$.

Set w_0 to be the permutation reversing $1, 2, \dots, n$

Theorem (B.-Gaetz 2025+)

If $\Gamma_{u,v}$ and $\Gamma_{u',v'}$ are isomorphic Bruhat graphs such that $[q]P_{u,v} = 0$ or $[q]P_{vw_0, uw_0} = 0$, then $R_{u,v} = R_{u',v'}$.

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If $\Gamma_{u,v}$ is isomorphic to $\Gamma_{e,v'}$, then $[q]P_{vw_0, uw_0} = 0$.

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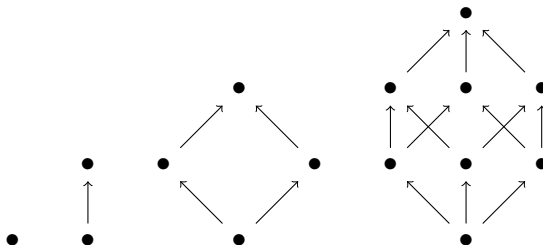
Machine learning Kazhdan–Lusztig polynomials

Blundell–Buesing–Davies–Veličković (from Google DeepMind) and Geordie Williamson trained a neural network to guess Kazhdan–Lusztig polynomials, given the poset.

- They asked the neural net which substructures of $\Gamma_{u,v}$ affected its answer a lot.
- It told* them about **hypercube decompositions**.
- They conjectured a recurrence for the $P_{u,v}(q)$ using hypercube decompositions which would imply the Combinatorial Invariance Conjecture.

Hypercubes

A **hypercube** is a directed graph that is the poset diagram of a Boolean lattice



Hypercube decompositions

Definition (Blundell–Buesing–Davies–Velickovic–Williamson 2022)

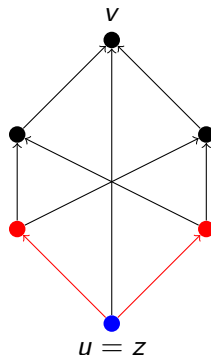
A **hypercube decomposition** of $\Gamma_{u,v}$ consists of:

- a subgraph $I = \Gamma_{u,z}$ and,
- for each $x \in I$ and $y \notin I$, the set of hypercubes $\mathcal{H}_x^y$ with bottom vertex x , top vertex y , and no edges in I

The hypercubes must satisfy:

- An existence condition
- A uniqueness condition: No two hypercubes have the same bottom edges

Non-example



Hypercube decompositions

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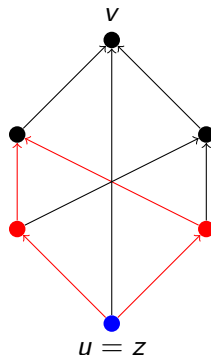
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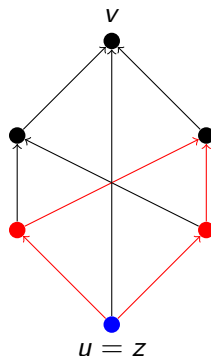
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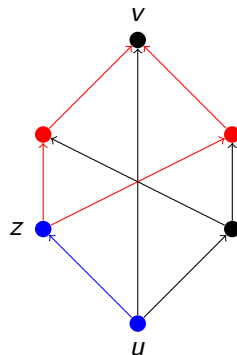
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Hypercube decompositions

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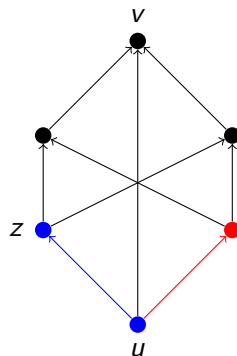
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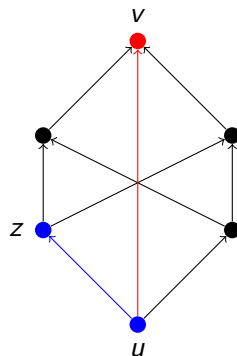
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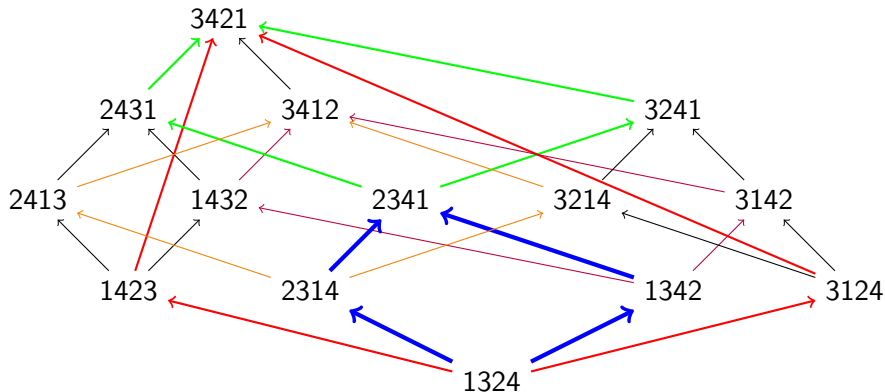
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Example



A bigger example



The BBDVW conjecture

Let $\text{rk}(H)$ denote the rank of a hypercube H , so that $|H| = 2^{\text{rk}(H)}$

Conjecture (Blundell–Buesing–Davies–Veličković–Williamson 2022)

Let I be any hypercube decomposition of $\Gamma_{u,v}$. Then

$$q^{\ell(u,v)} P_{u,v} \left(\frac{1}{q} \right) - P_{u,v} = \sum_{x \in I \setminus \{u\}} R_{u,x} P_{x,v} - q^{\ell(u,v)} \sum_{\substack{y \notin I \\ H \in \mathcal{H}_u^y}} (q^{-1} - 1)^{\text{rk}(H)} q^{-|\Lambda(H)|} P_{y,v} \left(\frac{1}{q} \right).$$

Has a natural geometric interpretation in this form

Proved by BBDVW for **standard hypercube decompositions**

Theorem

Theorem (B.–Gaetz 2025)

The BBDVW recursion holds for any hypercube decomposition of a Bruhat graph with the form $\Gamma_{e,v}$.

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These are the only cases of the BBDVW conjecture proven so far

A simplification: $\tilde{R}$ -polynomials

The BBDVW formula involves R -polynomials and P -polynomials and is very complicated. Is there a simpler way?

Definition (Brenti)

There is a unique family of polynomials $\tilde{R}_{u,v}(q) \in \mathbb{N}[q]$ such that

$$q^{\frac{\ell(v)-\ell(u)}{2}} \tilde{R}_{u,v}(q^{\frac{1}{2}} - q^{-\frac{1}{2}}) = R_{u,v}(q).$$

$$\begin{aligned} \text{Examples: } R_{u,v} &= (q-1)^5 & \rightsquigarrow & \tilde{R}_{u,v} = q^5 \\ R_{u,v} &= (q-1)^3 + q(q-1) & \rightsquigarrow & \tilde{R}_{u,v} = q^3 + q \end{aligned}$$

Equivalent form of Combinatorial Invariance Conjecture

If $\Gamma_{u,v} \cong \Gamma_{u',v'}$, then $\tilde{R}_{u,v}(q) = \tilde{R}_{u',v'}(q)$.

A conjecture

Conjecture (B.–Gaetz 2023, B.–Gaetz 2025)

Let $(I, \mathcal{H})$ be any hypercube decomposition of $\Gamma_{u,v}$ which satisfies a “numerical criterion”. Then

$$\sum_{\substack{y \notin I \\ H \in \mathcal{H}_u^y}} (-q)^{\text{rk}(H)} \tilde{R}_{y,v} = 0$$

and

$$\tilde{R}_{u,v} = \sum_{x \in I} \sum_{H \in \mathcal{H}_x^v} q^{\text{rk}(H)} \tilde{R}_{u,x}.$$

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$$\sum_{\substack{y \notin I \\ H \in \mathcal{H}_u^y}} (-q)^{\text{rk}(H)} \widetilde{R}_{y,v} = 0$$

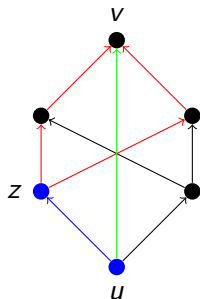
and

$$\widetilde{R}_{u,v} = \sum_{x \in I} \sum_{H \in \mathcal{H}_x^v} q^{\text{rk}(H)} \widetilde{R}_{u,x}.$$

Either formula would imply the combinatorial invariance conjecture!

Example

$$\begin{aligned}
 \tilde{R}_{u,v} &= \sum_{x \in I} \sum_{H \in \mathcal{H}_x^v} q^{\text{rk}(H)} \tilde{R}_{u,x} = \textcolor{green}{q} \tilde{R}_{u,u} + \textcolor{red}{q}^2 \tilde{R}_{u,z} \\
 &= \textcolor{green}{q} \cdot \textcolor{blue}{1} + \textcolor{red}{q}^2 \cdot \textcolor{blue}{q} = q + q^3 = \tilde{R}_{u,v}.
 \end{aligned}$$



- 1 Kazhdan–Lusztig polynomials
- 2 The Bruhat graph
- 3 The Combinatorial Invariance Conjecture
- 4 Hypercube decompositions
- 5 Coefficient of q

The Gabber–Joseph conjecture for S_n

To each u and v there is a topological space $\mathcal{R}_{u,v}$ called a **Richardson variety** with dimension $\ell(u, v)$

Gabber–Joseph Conjecture (1981)

The R -polynomial is given by

$$R_{u,v}(q) = \sum_i (-1)^{\ell(u,v)-i} q^i \dim H^i(\mathcal{R}_{u,v}; \mathbb{C}).$$

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Theorem (Boe 1992)

The Gabber–Joseph conjecture is false: there exist $u, v \in S_6$ such that the coefficients of $R_{u,v}$ do not alternate in sign

The coefficient of q

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The R -polynomial is given by

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Theorem (B.–Gaetz–Lam 2026)

Let $u, v \in S_n$. Then

$$[q]R_{u,v} = (-1)^{\ell(u,v)-1} \dim H^1(\mathcal{R}_{u,v}; \mathbb{C})$$

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Can we use this to prove combinatorial invariance for $[q]R_{u,v}$?

From cohomology to clusters to coefficients

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Cluster algebras also tell us about special structures in $\Gamma_{u,v}$ called **supporting chains**

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We can compute $\dim H^1(\mathcal{R}_{u,v}; \mathbb{C})$ by counting supporting chains in $\Gamma_{u,v}$!

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Theorem (Patimo 2021, B.–Gaetz–Lam 2026)

If $\Gamma_{u,v} \cong \Gamma_{u',v'}$, then $[q]R_{u,v} = [q]R_{u',v'}$ and $[q]P_{u,v} = [q]P_{u',v'}$.

The full conjecture

So far, we have discussed the CIC for S_n .

S_n is an example of a **Coxeter group**. These are groups with a special presentation.

For any two elements u, v in a Coxeter group W , there is a Bruhat graph $\Gamma_{u,v}$ and polynomials $P_{u,v}$ and $R_{u,v}$.

Full Combinatorial Invariance Conjecture

Let u and v be elements of any Coxeter group W , and let u' and v' be elements of any Coxeter group W' . If $\Gamma_{u,v} \cong \Gamma_{u',v'}$, then $P_{u,v} = P_{u',v'}$.

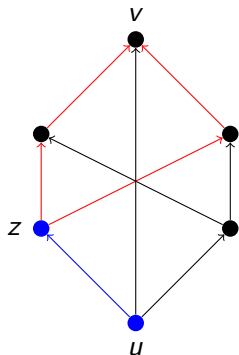
The coefficient of q in full generality

Theorem (B.–Gaetz–Lam 2026)

Let u and v be elements of any Coxeter group W , and let u' and v' be elements of any Coxeter group W' . If $\Gamma_{u,v} \cong \Gamma_{u',v'}$, then $[q]R_{u,v} = [q]R_{u',v'}$ and $[q]P_{u,v} = [q]P_{u',v'}$.

These are the only (non-trivial) coefficients of either polynomial known to be fully combinatorially invariant.

Thank you!



Hypercube decompositions

Definition (Blundell–Buesing–Davies–Velickovic–Williamson 2022)

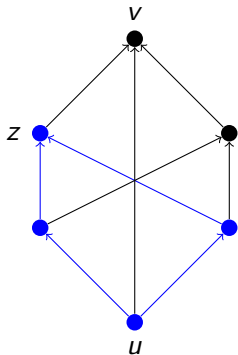
A subgraph $I = \Gamma_{u,z} \subset \Gamma_{u,v}$ forms a **hypercube decomposition** of $\Gamma_{u,v}$ if:

- (1) I is **diamond-closed** inside $\Gamma_{u,v}$.
- (2) For every $x \in I$, let $\mathcal{Y}_x = \{y \mid x \rightarrow y, y \notin I\}$ and let $\mathcal{A}_x$ denote the antichains of $\mathcal{Y}_x$ under Bruhat order. We require that every $A \in \mathcal{A}_x$ spans a unique hypercube in $\Gamma_{u,v}$.

A is an **antichain** if there is no path between any two elements of A

Non-example of hypercube decomposition

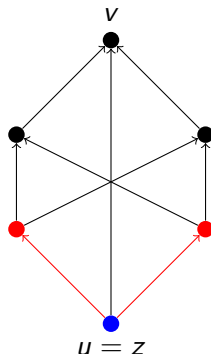
(1) I is **diamond-closed** inside $[u, v]$.



$I = \Gamma_{uz}$ not diamond-closed

Non-example of hypercube decomposition

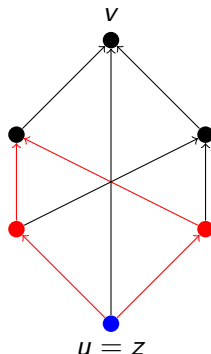
- (2) For every $x \in I$, let $\mathcal{Y}_x = \{y \mid x \rightarrow y, y \notin I\}$ and let $\mathcal{A}_x$ denote the antichains of $\mathcal{Y}_x$ under Bruhat order. We require that every $A \in \mathcal{A}_x$ spans a unique hypercube in $\Gamma_{u,v}$.



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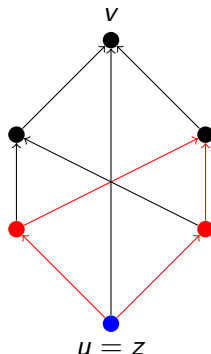
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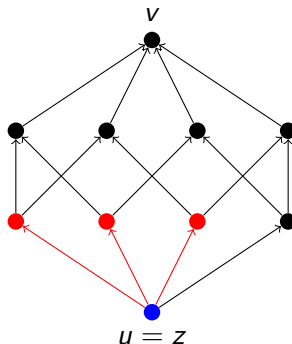
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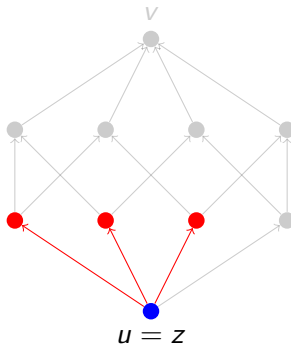
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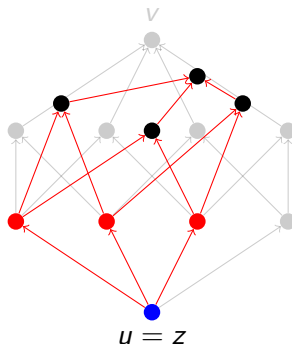
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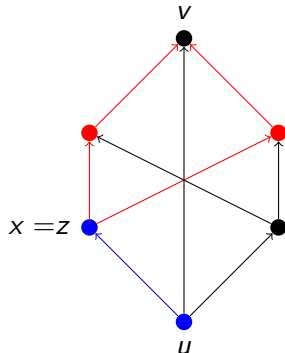
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Example of hypercube decomposition

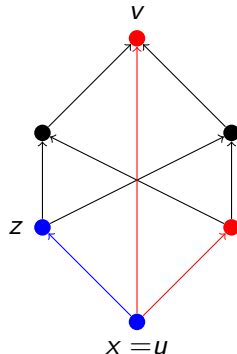
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$I = \Gamma_{u,z}$ is diamond-closed and hypercubes are unique

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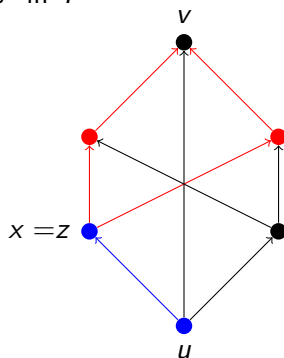


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The hypercube map

Let $\mathcal{I}$ is a hypercube decomposition and $Y \subseteq \mathcal{Y}_x$

- $\theta_x(Y) :=$ the top of the hypercube spanned by the antichain of maximal elements¹ in Y



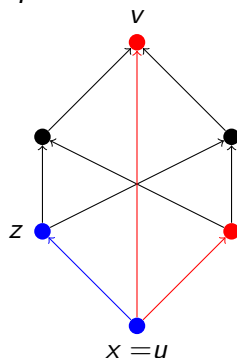
$$\theta_x(\mathcal{Y}_x) = v$$

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The numerical criterion

Let I be a hypercube decomposition of $[u, v]$.

Definition

We say I satisfies the **numerical criterion** (at u) if, for all $Y \in \mathcal{A}_u$,

$$\ell(u, \theta_u(Y)) = 2 |\{x \in \mathcal{Y}_u \mid \exists y \in Y, x \leq y\}| - |Y|.$$

Lemma

If I satisfies the numerical criterion, then

$$\sum_{A \in \mathcal{A}_u} (-1)^{|A|} q^{|A|} \tilde{R}_{\theta(A), v} = 0$$

implies the BBDVW conjecture for I .

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implies the BBDVW conjecture for I .

Remark: sometimes $\sum_{A \in \mathcal{A}_u} (-1)^{|A|} q^{|A|} \tilde{R}_{\theta(A), v}$ is nonzero!

The theorem

Theorem (B.–Gaetz 24+)

If I is a hypercube decomposition of an interval of the form $[e, v]$ (a **lower interval**), then I satisfies the numerical criterion and

$$\sum_{A \in \mathcal{A}_u} (-1)^{|A|} q^{|A|} \widetilde{R}_{\theta(A), v} = 0.$$

In particular, the BBDVW conjecture is true for lower intervals.

The theorems

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$$\sum_{A \in \mathcal{A}_e} (-1)^{|A|} q^{|A|} \widetilde{R}_{\theta(A), v} = 0.$$

In particular, the BBDVW conjecture is true for lower intervals.

Theorem (B.–Gaetz 23)

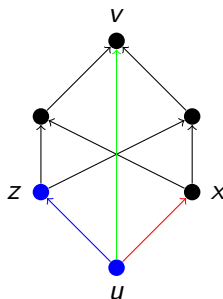
If I is a hypercube decomposition of a lower interval $[e, v]$, then

$$\widetilde{R}_{ev} = \sum_{x \in I} \sum_{\substack{A \in \mathcal{A}_x \\ \theta_x(A) = v}} q^{|A|} \widetilde{R}_{ex}.$$

Example of BBDVW

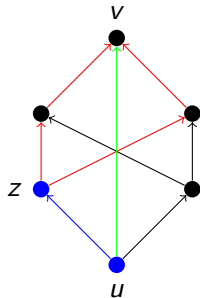
$$\sum_{A \in \mathcal{A}_e} (-1)^{|A|} q^{|A|} \tilde{R}_{\theta(A), v} = \tilde{R}_{u, v} - \textcolor{green}{q} \tilde{R}_{vv} - \textcolor{red}{q} \tilde{R}_{xv}(q)$$

$$= q + q^3 - \textcolor{green}{q} \cdot 1 - \textcolor{red}{q} \cdot q^2 = 0.$$



Example of BG23

$$\begin{aligned}
 \tilde{R}_{u,v} &= \sum_{x \in I} \sum_{\substack{A \in \mathcal{A}_x \\ \theta_x(A) = v}} q^{|A|} \tilde{R}_{u,x} = \textcolor{green}{q} \tilde{R}_{u,u} + \textcolor{red}{q}^2 \tilde{R}_{u,z} \\
 &= \textcolor{green}{q} \cdot \textcolor{blue}{1} + \textcolor{red}{q}^2 \cdot \textcolor{blue}{q} = q + q^3 = \tilde{R}_{u,v}.
 \end{aligned}$$



The conjecture

Conjecture

Let I be any hypercube decomposition of $\Gamma_{u,v}$ which satisfies the numerical criterion. Then

$$\sum_{A \in \mathcal{A}_u} (-1)^{|A|} q^{|A|} \tilde{R}_{\theta(A),v} = 0$$

and

$$\tilde{R}_{u,v} = \sum_{x \in I} \sum_{\substack{A \in \mathcal{A}_x \\ \theta_x(A)=v}} q^{|A|} \tilde{R}_{u,x}.$$

The proof

$\tilde{R}$ -polynomials and reflection orders

Definition

A **reflection order** is a total order $\prec$ on transpositions so that for all $1 \leq a \leq b \leq c \leq n$,

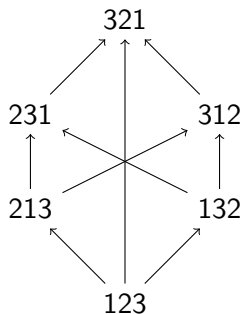
$$(ab) \prec (ac) \prec (bc) \quad \text{or} \quad (bc) \prec (ac) \prec (ab).$$

Theorem (Dyer 1987)

Fix a reflection order $\prec$. The coefficient of q^k in $\tilde{R}_{u,v}$ is the number of **increasing paths** from u to v of length k .

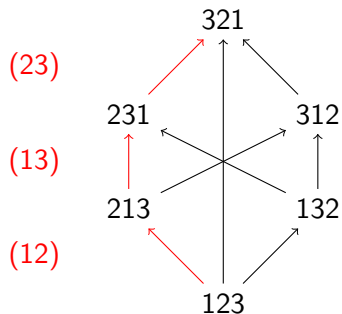
$\tilde{R}$ -polynomial example

$$(12) \prec (13) \prec (23)$$



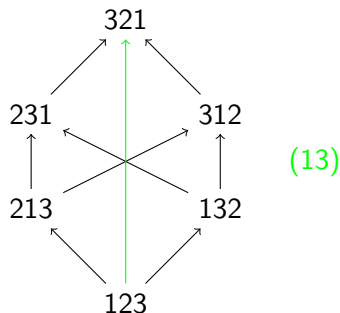
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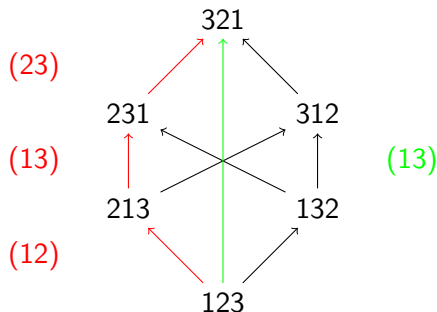
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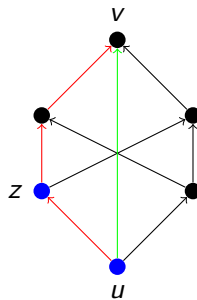
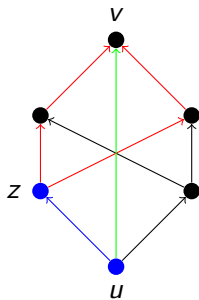
$$(12) \prec (13) \prec (23)$$



$$\tilde{R}_{123,321} = q^3 + q^1$$

Proof idea

$$\sum_{x \in I} \sum_{\substack{A \in \mathcal{A}_x \\ \theta_x(A) = v}} q^{|A|} \tilde{R}_{u,x} = \tilde{R}_{u,v},$$



The lemma

A rearrangement

BBDVW conjecture equivalent to

$$\sum_{Y \subseteq \mathcal{Y}_u} q^{\ell(u, \theta(Y))} (q^{-1} - 1)^{|Y|} \sum_{y \in [u, v] \setminus I} R_{\theta(Y), y} P_{yv} = 0.$$

What if already

$$\sum_{Y \subseteq \mathcal{Y}_u} q^{\ell(u, \theta(Y))} (q^{-1} - 1)^{|Y|} R_{\theta(Y), y} = 0?$$

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$$\sum_{Y \subseteq \mathcal{Y}_u} q^{\ell(u, \theta(Y))} (q^{-1} - 1)^{|Y|} R_{\theta(Y), y} = 0?$$

Let's simplify even more.

Simplifying further

Lemma

Assume that I is a hypercube decomposition of $[u, v]$ satisfying the numerical criterion. Then

$$\sum_{Y \subseteq \mathcal{Y}_u} q^{\ell(u, \theta(Y))} (q^{-1} - 1)^{|Y|} R_{\theta(Y), y} = 0$$

if and only if

$$\sum_{Y \in \mathcal{A}_u} (-1)^{|Y|} q^{|Y|} \tilde{R}_{\theta(Y), y} = 0.$$